

Armaghan Sarvar

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SUMMARY Bioinformatics scientist specializing in machine learning for biological sequence modeling, computational genomics, and cancer epigenomics. Develops deep learning and statistical methods for sequence-to-function prediction, cfDNA methylation analysis, chromatin profiling, and single-cell genomics. Recent RECOMB 2026 work focused on molecular residual disease detection in triple-negative breast cancer and genotype-resolved modeling of G-quadruplex occupancy.

PROFESSIONAL **Bioinformatics Scientist** December 2023–Present
EXPERIENCE *Aparicio Lab, BC Cancer Research Centre, Vancouver*

- Developed **sequence-to-function deep learning models** to predict chromatin G-quadruplex occupancy and quantitative CUT&Tag signal directly from DNA sequence across cancer-relevant breast epithelial genotypes.
- Built multi-task learning workflows for genome-scale regulatory signal prediction, integrating DNA sequence, model-derived embeddings, genomic annotations, epigenetic features, and CpG modification context.
- Developed **Mod-squares**, a plasma cfDNA methylation framework for estimating tumour-derived signal and improving low-fraction molecular residual disease detection in triple-negative breast cancer.
- Designed **5mC/hmC-aware feature extraction** workflows for Biomodal six-letter sequencing data, enabling single-base-resolution analysis of methylation and hydroxymethylation patterns.
- Processed and analyzed **Illumina short-read sequencing** data across cancer genomics workflows, including alignment, signal extraction, quality control, and downstream statistical modeling.
- Analyzed **DLP+ single-cell DNA sequencing** data to identify copy number based predictors of drug response and characterize tumour genomic heterogeneity.
- Performed mutational signature analysis, CRISPR perturbation analysis, and visualization of genome-scale functional genomics datasets for collaborative cancer research projects.

Machine Learning Engineer and Consultant March–December 2023
Trauma Surgery Team, Vancouver General Hospital, Vancouver

- Improved predictive performance for clinical classification and regression problems using ensemble ML methods with hyperparameter optimization and cross-validation.
- Performed preprocessing, imputation, one-hot encoding, recursive feature elimination, class imbalance handling, and outlier detection in close collaboration with the medical team.

Graduate Research Assistant September 2021–December 2023
Bioinformatics Technology Lab, Michael Smith Genome Sciences Centre, BC Cancer Agency
Supervisor: Dr. Inanc Birol

- Designed novel algorithms and data structures for genome misassembly detection, including Stash, a stochastic tile hashing approach presented at ISMB 2022 and RECOMB 2023.
- Developed scalable computational genomics workflows for genome assembly analysis, sequence comparison, and algorithmic evaluation on large genomic datasets.

Bioinformatics Research Intern May–July 2022
Silent Genomes Project, BC Children's Hospital Research Institute, Vancouver

- Built long-read pipelines for **PacBio HiFi** and **Oxford Nanopore** data, including alignment, SNV/Indel calling, structural variant detection, methylation calling, and STR genotyping.
- Applied long-read sequencing workflows to intracranial aneurysm linkage analysis in an Indige-

nous family as part of a clinical genomics research collaboration.

Undergraduate Research Assistant

March 2020–February 2021

Amirkabir University of Technology Supervisor: Dr. Maryam Amirmazlaghani

- Developed a wavelet domain defense against adversarial attacks on deep learning image classifiers, published in *Applied Intelligence*.

EDUCATION

University of British Columbia

2021–2023

M.Sc. Bioinformatics GPA: 95.3/100 Thesis: *Genome Misassembly Detection using Data Structure and Algorithm Design*

Amirkabir University of Technology

2016–2020

B.Sc. Computer Engineering GPA: 3.75/4 Thesis: *Defending Deep Learning Models against Adversarial Attacks*

SELECTED

PUBLICATIONS

- **Genome misassembly detection using Stash: A data structure based on stochastic tile hashing**

Armaghan Sarvar, Lauren Coombe, and Inanc Birol.

PLOS ONE, 2026

AND

PRESENTATIONS

- **The UV-potentiated mutational signature of the clinical-stage G-quadruplex binder CX5461**

Elena Zaikova, Damian B. Yap, Armaghan Sarvar, et al.

bioRxiv, 2026. Preprint

- **Mod-squares detects low-fraction residual disease and predicts recurrence from plasma methylation in triple-negative breast cancer**

Armaghan Sarvar et al.

RECOMB-CCB, 2026. Contributed talk

- **Multi-task modeling of genotype-resolved chromatin G-quadruplex occupancy and quantitative CUT&Tag signal from DNA sequence**

Armaghan Sarvar et al.

RECOMB and RECOMB-CCB, 2026. Poster presentation

- **Misassembly detection using Stash: A data structure based on stochastic tile hashing**

Armaghan Sarvar, Lauren Coombe, René Warren, and Inanc Birol.

RECOMB, 2023.

Contributed talk

- **Stash: A data structure based on stochastic tile hashing**

Armaghan Sarvar, Lauren Coombe, René Warren, and Inanc Birol.

ISMB, 2022. Contributed talk

talk

- **Defense against adversarial examples based on wavelet-domain analysis**

Armaghan Sarvar and Maryam Amirmazlaghani.

Applied Intelligence, 2022

SKILLS

Languages:

Python, R, Bash, Java, C/C++, Matlab, Typst

ML & DL:

PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SciPy, Hugging Face

Modeling:

Sequence-to-function modeling, multi-task learning, embeddings, attention, CNNs, transformers, autoencoders, ensemble methods, hyperparameter optimization

Bioinformatics:

Snakemake, Nextflow, GATK, samtools, bcftools, minimap2, BWA, DESeq2, HMMcopy, bedtools, Mosdepth, IGV, Biopython, pysam, pyBigWig, methylKit, QUAST

Sequencing & Assays:

Illumina short-read sequencing, Oxford Nanopore, PacBio HiFi, DLP+, CUT&Tag, Biomodal six-letter sequencing, cfDNA methylation, single-cell DNA sequencing

Data & Viz:

Matplotlib, Seaborn, Shiny, Power BI, Jupyter, dimensionality reduction, clustering

Infrastructure:

Git, Linux, HPC, Slurm, Conda, Singularity, Docker, MySQL, PostgreSQL, Azure Blob Storage

SELECTED PROJECTS	● Multi-task DNA Sequence-to-Function Modeling for Chromatin G4 Biology: Developed a deep learning framework to jointly predict genotype-resolved chromatin G-quadruplex occupancy and quantitative CUT&Tag signal from DNA sequence, using model embeddings and genomic annotations to study regulatory G4 subtypes.		
	● Foundation-Model-Inspired Regulatory Genomics: Adapted concepts from long-context sequence-to-function models such as Enformer and Borzoi to cancer epigenomics tasks, including quantitative regulatory signal prediction, embedding analysis, and feature attribution across genomic loci.		
	● Plasma cfDNA Methylation Deconvolution for Molecular Residual Disease: Built Mod-squares, a CpG-modification-based framework to estimate tumour-derived signal from plasma cfDNA and improve recurrence-associated ctDNA detection in triple-negative breast cancer.		
	● 5mC/hmC-Aware Feature Engineering for Six-Letter Sequencing: Designed single-base-resolution methylation and hydroxymethylation feature extraction workflows from Biomodal sequencing data for downstream statistical modeling and epigenomic characterization.		
	● Scalable Cancer Genomics Pipelines: Built reproducible workflows for Illumina, Oxford Nanopore, PacBio HiFi, DLP+, CUT&Tag, and methylation sequencing data, including QC, alignment, signal extraction, variant and methylation analysis, and visualization.		
TEACHING AND SERVICE	● Transferable ML Methods for Biological Sequence Modeling: Applied embeddings, multi-task objectives, attention-based architectures, and quantitative sequence-to-function prediction to genome-scale biological data, with relevance to protein sequence and function modeling.		
	Mentor, UBC National Canadian Medical Datathon Judge, BC Cancer Research Day Judge, UBC BiGSAT (Bioinformatics and Genome Science & Technology) Conference Teaching Assistant, Master of Data Science, UBC Communication in Data Science; Data Visualization II; Computing Platforms for Data Science Organizing Committee, Stemcelltalks Event Teaching Assistant, Amirkabir University of Technology Data Mining; Fundamentals of Image Processing; Linear Algebra	2025, 2026	
		2024, 2025	
		2024, 2025	
		2022–2023	
		2022	
PROFESSIONAL DEVELOPMENT	2019–2020		
	● Generative AI with Large Language Models, DeepLearning.AI / AWS		
	● AI for Medicine Specialization, Stanford University		
	● Applied Machine Learning in Python, University of Michigan		
	● Neural Networks and Deep Learning, Stanford University		
LANGUAGES	● Project Management Fundamentals, UBC Sauder School of Business		
	● Conference Attendance: MICCAI (2023), NeurIPS (2024), ICML (2025), RECOMB (2023, 2026)		
	● Persian: Native English: Fluent Turkish: Fluent Arabic: Familiar		